

Summary Sheet

When running a sports league, a bad schedule can completely break a competition, whether it be by creating match-ups between teams with a large skill disparity, or overworking teams during the league by giving them too many consecutive matches. Hence, we have found that the creation of a model that creates a **“fair and effective”** schedule is imperative to making the Global Sports League (GSL) a smashing success.

We redefine what the IMMC calls a “fair and effective” schedule as one that gives all participating countries an **equal opportunity of achieving victory** by minimising the impact of any external factors on winning. Furthermore, to accommodate modern sensibilities, we have ensured that the participating countries are spread across the entire world, promoting diversity in the GSL, and that our schedule **maximises** the **economic** and **environmental sustainability** of the GSL through carbon footprint and waste minimisation.

In picking our first sport for the GSL’s debut, we picked **volleyball**, as relative to other sports, it is unaffected by external factors affecting **fairness** and has a low carbon footprint, making it more fair and sustainable. We employed **K-Means Clustering** to group countries into divisions based on location, financial strength and competitive strength in order to ensure a diverse, yet even spread of countries. Then, to narrow down the participating countries from each division, we **qualitatively assessed** the existing sports infrastructure and facilities of each country, as well as their competitive viability in a volleyball tournament.

Our scheduling model uses the **Haversine Formula** to minimize the distance each participating country travels through using their latitude and longitude to calculate the shortest distance between countries. Our model also employs the use of a **Double Round-Robin Tournament** system, which ensures the creation of a fair and feasible schedule by forcing each team in a division to face each other twice. However, as there is an existing trade-off between geographic diversity and fairness of match-ups, we also utilised an **Elo-based fairness model**, which ranks teams based on their performance in each match over time and factors in the strength of their opponents into the ranking, showing off each team’s true competitive strength.

To expand our model to accommodate for more teams, we used k-means clustering again to determine the **additional** potential participating countries and then manually evaluated which ones had the **best volleyball infrastructure** in each division region to reduce travel time. After the addition of the four new teams, the **number of games per week** in the 36 week GSL season **increased** from 7 to 10, which would cause the schedule to be **overcrowded**. However, the increase in teams also increases the number of games played, by the law of large numbers, causes the observed performance of each team to converge to their true skill level.

Following that, we adjusted our current scheduling model to accommodate a wider range of sports with different rulesets and environments than volleyball, through accounting for changes in how we provide elo for different sports, as well as changing the structure of the league from round-robin to other formats. We also employed a **Margin of Victory Adjustment Formula** that will change the Elo rating system to scale rating changes based on the decisiveness of a win to improve its rating accuracy. Furthermore, we used **draw probability** to adjust the Elo model for sports involving draws.

Finally, we carried out a **Sensitivity Analysis** by using a **Sobol Sensitivity Test** and a **Monte Carlo Simulation**, allowing us to find out what factors affect our model the most. We also did a **Validity Analysis**, wherein we discovered how efficient our model truly performs in a real world context.

Letter to the decision makers

Dear International Multi-Continental Matchmaking Committee heads,

It is our greatest pleasure to state that we were able to successfully create a mathematical model that can make schedules which completely align with the goals of the IMMC. Our scheduling model fully recognises the importance of fair match-ups between different teams, and is tailored to ensure that each team has an equal chance of winning the GSL. We initially created the model with volleyball in mind as it is a sport that is both sustainable and fair.

In the pursuit of geographical diversity, the model splits all countries into separate divisions based on their geographical location, economic strength, and previous competitive sports history. This allows us to have an even distribution of countries of differing backgrounds, and ensures that each division will consist of teams of a similar skill level or economic background. This will lessen the effects of a skill and resource imbalance between countries.

This model also creates match-ups in the schedule based on travel distance and competitive strength to ensure competitive equity and to minimise travel distance. This leads to the creation of a schedule consisting of fair match-ups, and in tandem, also lessens how much each team needs to travel, reducing the effects of player travel fatigue, and reducing the fossil fuel needed to travel from one place to another, increasing the sustainability of the event. Our model ensures fairness through forcing each team in every division to play against each other two times, giving them more chances to show their true potential. Moreover, our model balances the amount of matches each team plays in their home country and overseas, once again reducing the extent of travel fatigue and reducing any advantage a team may have from playing in their home country.

However, since we are prioritising the sustainability of the scheduled GSL league through cutting down on travel distance, we had to at first sacrifice a bit of our focus on fairness in our model. To counteract this, we implemented a system that ranks our teams based on their performance in the competition and proven strength by weighing the matches' predicted winners against the actual winners of the match. The scheduling model then takes these rankings into account and schedules future matches based off of that.

On top of accommodating for this range of factors, our model can be applied to a myriad of sports with minor adjustment depending on their rules and location. Moreover, it can also be expanded through the addition of more teams, with a few caveats as some teams may end up having more close together matches than others, and create a crowded schedule if sticking to the 36-week season system proposed by the IMMC.

In summary, our model is able to create practical and balanced schedules that take into account external factors that may affect player performance, ensures geographical diversity through picking countries from all across the world for each division, and also promotes sustainability through minimising our carbon footprint.

Thank you for your time and consideration.

Yours faithfully,

Team []

Initial 20-Team GSL Schedule

Week	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
1	JPN vs. CHN	IRN vs. ITA	ARG vs. SRB	TUN vs. JPN	IRN vs. AUS	SRB vs. MEX	CAN vs. AUS
2	CAN vs. NZL	CHN vs. JPN	ITA vs. IRN	SRB vs. ARG	TUN vs. CAN	IRN vs. SRB	SRB vs. NZL
3	SRB vs. CAN	MEX vs. AUS	JPN vs. AUS	POL vs. ITA	RUS vs. FRA	JPN vs. IRN	IRN vs. CUB
4	IRN vs. NZL	FRA vs. GER	MEX vs. ARG	AUS vs. JPN	ITA vs. POL	FRA vs. RUS	JPN vs. GER
5	JPN vs. USA	IRN vs. MEX	FRA vs. MEX	MEX vs. BRA	JPN vs. NZL	USA vs. CAN	RUS vs. SRB
6	SRB vs. RUS	JPN vs. CAN	POL vs. CUB	FRA vs. AUS	MEX vs. NZL	NZL vs. JPN	CAN vs. USA
7	USA vs. MEX	FRA vs. SRB	JPN vs. SRB	POL vs. MEX	FRA vs. CAN	CUB vs. ARG	JPN vs. NED
8	NED vs. JPN	MEX vs. USA	SRB vs. FRA	JPN vs. ARG	POL vs. ARG	FRA vs. USA	CUB vs. NZL
9	CUB vs. BRA	CHN vs. AUS	USA vs. CUB	EGY vs. AUS	JPN vs. ITA	POL vs. SRB	FRA vs. CUB
10	FRA vs. NZL	CUB vs. AUS	AUS vs. CHN	CUB vs. USA	EGY vs. CHN	JPN vs. CUB	POL vs. FRA
11	POL vs. RUS	FRA vs. NED	BRA vs. AUS	CHN vs. NZL	USA vs. GER	EGY vs. CAN	JPN vs. FRA
12	JPN vs. BRA	POL vs. NED	RUS vs. NZL	BRA vs. NZL	NZL vs. CHN	GER vs. USA	EGY vs. ARG
13	EGY vs. RUS	JPN vs. MEX	POL vs. CAN	RUS vs. CUB	ARG vs. NZL	CHN vs. NED	CAN vs. MEX
14	MEX vs. CAN	EGY vs. JPN	JPN vs. POL	POL vs. USA	RUS vs. USA	ARG vs. AUS	NED vs. CHN
15	AUS vs. NZL	CAN vs. CUB	EGY vs. NZL	JPN vs. RUS	POL vs. NZL	RUS vs. GER	
16		NZL vs. AUS	CUB vs. CAN	EGY vs. CUB	CHN vs. ITA	POL vs. GER	RUS vs. NED
17	RUS vs. AUS		AUS vs. NED	CAN vs. GER	EGY vs. NED	CHN vs. IRN	POL vs. AUS
18	POL vs. BRA	RUS vs. CAN		NED vs. AUS	GER vs. CAN	EGY vs. USA	CHN vs. BRA
19	CHN vs. RUS	ITA vs. SRB	RUS vs. MEX		NZL vs. NED	MEX vs. CUB	EGY vs. BRA
20	EGY vs. FRA	CHN vs. USA	ITA vs. MEX	GER vs. BRA		NED vs. NZL	CUB vs. MEX
21	MEX vs. GER	EGY vs. MEX	CHN vs. ARG	ITA vs. NED	GER vs. NZL		EGY vs. TUN
22	TUN vs. EGY	GER vs. MEX	EGY vs. SRB	CHN vs. SRB	ITA vs. GER	GER vs. NED	
23		EGY vs. IRN	CUB vs. GER	EGY vs. GER	CHN vs. MEX	ITA vs. BRA	GER vs. ARG
24	GER vs. AUS		IRN vs. EGY	GER vs. CUB	TUN vs. CHN	CHN vs. GER	ITA vs. ARG
25	ITA vs. CAN	NED vs. BRA		EGY vs. POL	BRA vs. ARG	TUN vs. FRA	CHN vs. CUB
26	CHN vs. FRA	ITA vs. NZL	NED vs. MEX		POL vs. EGY	ARG vs. BRA	TUN vs. CUB
27	TUN vs. ARG	CHN vs. CAN	ITA vs. USA	NED vs. CUB		EGY vs. ITA	BRA vs. RUS
28	RUS vs. BRA	TUN vs. BRA	CHN vs. POL	ITA vs. FRA	NED vs. USA		ITA vs. EGY
29	TUN vs. IRN	BRA vs. FRA	TUN vs. USA	IRN vs. NED	ITA vs. AUS	NED vs. CAN	
30		IRN vs. TUN	FRA vs. BRA	TUN vs. NED	IRN vs. USA	ITA vs. RUS	NED vs. ARG
31	USA vs. BRA		TUN vs. POL	BRA vs. SRB	TUN vs. MEX	IRN vs. BRA	ITA vs. CUB
32	SRB vs. NED	USA vs. AUS		POL vs. TUN	SRB vs. BRA	TUN vs. AUS	IRN vs. FRA
33	IRN vs. GER	SRB vs. CUB	USA vs. ARG		TUN vs. ITA	ARG vs. RUS	TUN vs. SRB
34	TUN vs. RUS	IRN vs. CAN	SRB vs. AUS	USA vs. NZL		ITA vs. TUN	RUS vs. ARG
35	ARG vs. FRA	TUN vs. NZL	IRN vs. RUS	SRB vs. GER	CAN vs. BRA		IRN vs. POL
36	POL vs. IRN	FRA vs. ARG	TUN vs. GER	IRN vs. ARG	SRB vs. USA	CAN vs. ARG	

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1 Introduction

1.1 Background

When running and organising any sports league, whether it be on a national or global scale, a schedule can make or break the competition. Proper scheduling is imperative to ensuring that any sports league is compelling to its viewers, as a poor schedule can result in unfair matchups, evidenced by the Belgian Soccer League Debacle [43], where many weaker teams were forced to compete against the stronger ones early on, leading to many teams dropping out, and for viewership to decline.

In the modern age, other factors are also taken into account when scheduling events, such as geographical diversity and sustainability, which affect the spread of matches in the sports league. Thus, we have found that we need to create a mathematical model that can create proper schedules for the Global Sports League (GSL) that not only prioritises efficiency but also the other important areas of interest that the International Multi-Continental Matchmaking Committee (IMMC) has put in place for us.

1.2 Problem Restatement and Analysis

In creating a schedule for the upcoming GSL, we have redefined the IMMC's request for a "fair and effective" schedule as one that gives all participants an equal opportunity of winning by minimising the effects of external factors. This allows all participating teams to be able to win through their own effort and not through outside effects. Moreover, we also factor in their requests for "geographic diversity" and "economic and environmental sustainability" as ensuring an even spread of countries across the world, and ensuring minimal wastage of resources.

1.3 Assumptions

Assumption 1: All teams are composed of only male team members.

We will assume that all teams participating in the GSL are composed solely of men and that there are no female divisions of the GSL.

Justification: As male sports are more popular than female sports [1], it is assumed that men's sports will naturally attract a greater audience, allowing for greater potential for global expansion than women's sports.

Assumption 2: All participants are professional athletes.

It is assumed that all participants are professionals who would fulfill the requirements set by the International Olympic Committee [2] to partake in the Olympics.

Justification: As GSL is a similarly large-scale international sporting event such as the Olympics, it is fair to expect that the athletes participating in the GSL are of a similar caliber to those who participate in the Olympics.

Assumption 3: All matches happen as scheduled without any delays or disqualifications.

It is assumed that any sudden delays (e.g. heavy traffic, flight delays, harsh weather) or rule breakages causing a disqualification will significantly disrupt the scheduled match-ups.

Justification: In most large-scale competitions, ample buffer time is present in between matches to account for such delays and hence will not cause much issue [41]. In addition, all participating athletes are professionals and are thus unlikely to partake in illicit activities that would incur disqualification.

Assumption 4: Matches never end in a draw.

It is assumed that every match ends with a clear winner and a clear loser.

Justification: In professional volleyball, games never end in draws [42]. Matches are played in a best-of-five format, where the first time to win three sets wins the match. To win each set, the team must win by at least two points. This system ensures that every match has a clear winner, with no possibility of a draw.

2 Task 1: Selection of Sports and Participating Teams

2.1 Initial Research on The Topic

In creating a model that will create a suitable schedule for the GSL, we would need to maximise our time and efficiency whilst minimising any harmful ecological and economic impacts. To do so, we have identified some key components contributing to environmental and economic sustainability to consider [9]:

Component	Explanation
Resource Efficiency	Using less energy and materials
Sustainable Innovation	Using and developing more environmentally friendly technology
Cost-Effectiveness	Maximising profitability while minimising waste
Circular Economy	Materials are circulated in the economy through recycling and reuse of waste

Figure 1: Key Factors in Environmental and Economic Sustainability

Moreover, to guarantee that all teams have an equal chance of winning, we want to evaluate the fairness of a sport and its schedule for different countries. To do so, we found some significant external factors that contribute to a greater likelihood of a sports team winning [3] as listed below :

Factors Affecting Victory in Sports	
Factors	Reasoning
Share of world income	Countries with wealth can allocate more funds to develop better sports facilities and coaching.
Population growth	A larger population causes a larger sports talent pool
Home advantage	Players playing in home territory have an advantage as they have more local support, less travel fatigue, and are more used to the climate
Distance to host country	Greater travel distance affects the extent of jet lag and unfamiliar climates, negatively impacting player performance

Figure 2: External Factors Influencing a Sports Team's Chances of Winning

2.2 Identifying a Sport Suited for Global Expansion

When picking a sport for our first schedule, we accounted for the sport’s fairness and sustainability, along with its global expansion potential, taking reference to its popularity and market saturation.

Hence, **Volleyball** was chosen as our first sport as it is a team sport that is played indoors, reducing climate-related effects of home advantage, as well as justifying Assumption 3 further, as the effects of weather are negligible to an indoor sport. It also has a low carbon footprint, making it environmentally sustainable (Figure 3). Moreover, volleyball requires minimal equipment to play [10], making it more widely accessible to countries with lower GDP, and making it more resource-efficient. Finally, volleyball already has an established international audience and player base (Figure 4), ensuring a strong foundation for participation and viewership, while still having more room to grow, making it prime for global expansion.

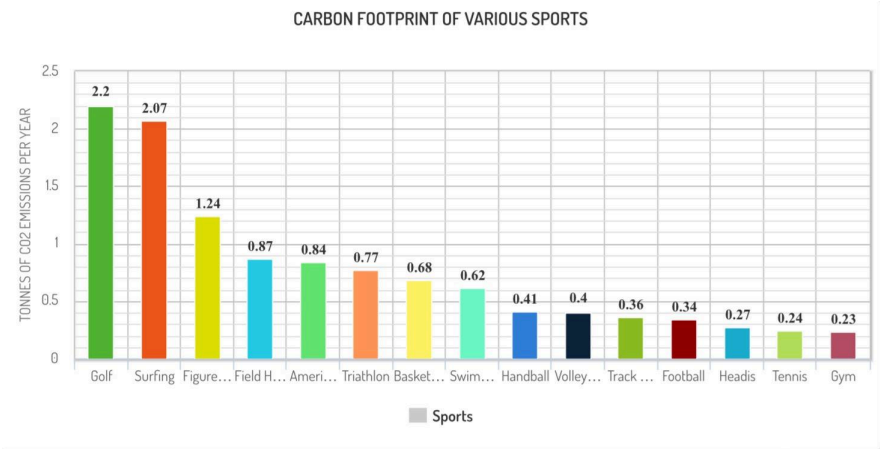


Figure 3: Carbon Footprint of Various Sports [44]

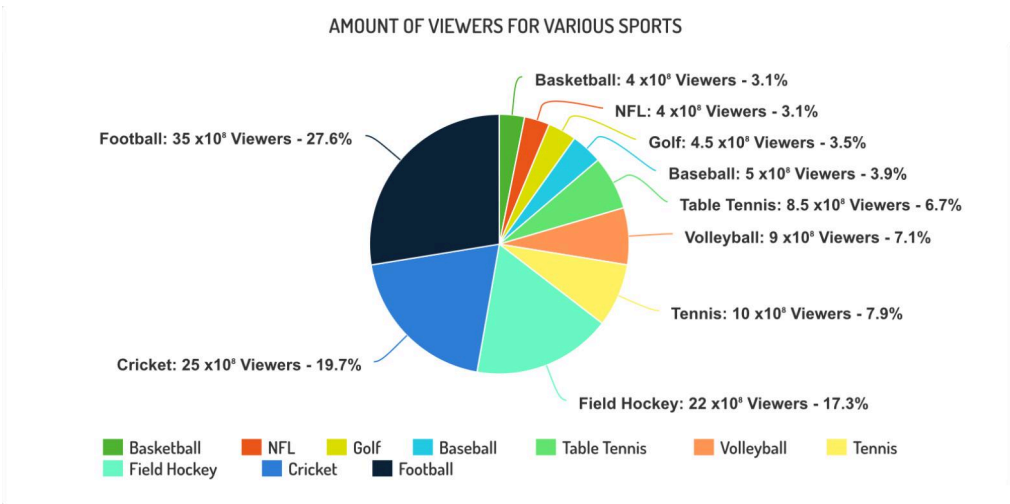


Figure 4: Viewer Base of Various Sports [45]

2.3 Division Formation and Structure

In creating a schedule for the GSL, we have decided to utilise a double round-robin system with divisions, wherein we will divide all 20 teams into 4 divisions with 5 teams each, based on their location, wealth and

previous volleyball experience. This system was used because it is one of the most commonly used tournament structures in scheduling [11], and ensures a fair and balanced schedule (refer to 3.2).

The selection of teams and grouping of divisions:

A set of key criteria guided the selection of teams and cities for the GSL to ensure geographic diversity, competitive equity, and logistical feasibility to align with IMMC's goals of global expansion and fair representation.

GSL Team Selection Criteria :

1. Geographic Diversity → to ensure a wide spread of countries
2. Volleyball Competitive Strength → to guarantee a fair and entertaining competition
3. Economic Strength → to ensure that no country is significantly wealthier
4. Existing sports infrastructure → to ensure that selected countries can host matches

In selecting what countries the participating teams will be coming from, we have utilized **K-Means Clustering** to determine what countries are viable choices for the GSL based on the GSL Team Selection Criterion, as well as **qualitative analysis** to determine what cities from each country are best suited for the GSL. We have done so to achieve a fair balance between geographical diversity and fairness between teams.

2.3.1 K-Means Clustering

K-Means Clustering is a model that takes unlabelled data and classifies it into a certain number of clusters based on the similarity of the data being considered. We have chosen to use K-Means clustering as it can select participating nations in each division based on geographical proximity, akin to how major sports leagues such as the National Football League [7] schedule their games, ensuring geographical representation of differing regions. Moreover, K-Means clustering allows us to factor in other variables such as wealth and competitive strength when grouping the clusters, ensuring that the teams in each division are equally distributed in terms of resources and competitive strength.

The objective of K-Means is to **minimize the total squared distance** between each team and its assigned cluster centroid:

$$\min \sum_{i=1}^k \sum_{x_j \in C_i} \|x_j - \mu_i\|^2$$

Where:

- k : numbers of clusters (divisions in GSL)
- x_j : team's feature factors (latitude, longitude, GDP per capita, Elo rating)
- C_i : cluster i (set of teams in the same division)
- μ_i : centroid of cluster C_i , computed as:

$$\mu_i = \frac{1}{|C_i|} \sum_{x_j \in C_i} x_j$$

Teams are assigned to the nearest cluster based on Euclidean distance, a measure of straight-line distance between two points:

$$d(x_j, \mu_i) = \sqrt{\sum_{m=1}^n (x_{jm} - \mu_{im})^2}$$

2.3.2 Applying the Elbow Method for Cluster Selection

To determine the optimal **number of clusters** in K-Means Clustering, we employed the **Elbow Method**, which analyzes the **Within-Cluster Sum of Squares (WCSS)**, a measurement of how close data points are to the centre of their cluster.:

$$WCSS = \sum_{i=1}^k \sum_{x_j \in C_i} \|x_j - \mu_i\|^2$$

We plotted WCSS against the number of clusters and found the “elbow point” at $k = 6$ (Figure 5), indicating the best trade-off between diversity and travel efficiency, allowing us to create divisions which are geographically diverse while minimising travel distances [17] [A.1].

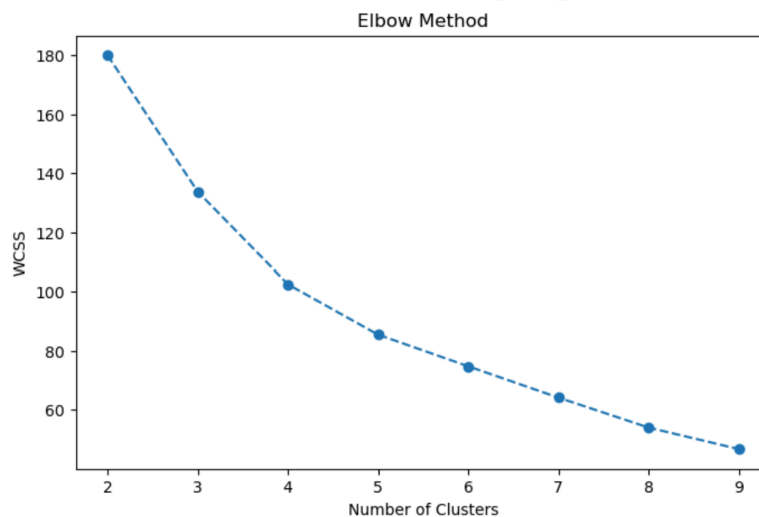
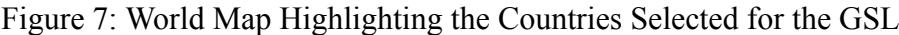


Figure 5: Elbow Method Diagram for the Optimal Number of Clusters

Next, to sort countries into divisions based on their latitude, longitude [4], GDP per capita [6] and previous volleyball world rankings [5], we used a K-Means clustering algorithm [A.1] to group our countries on a graph.



In creating a league schedule, we factored in all requests from the IMMC and external research [18] to create the GSL Sports Scheduling Framework (Figure 8), which comprises the main aspects of building a league schedule that we have determined.

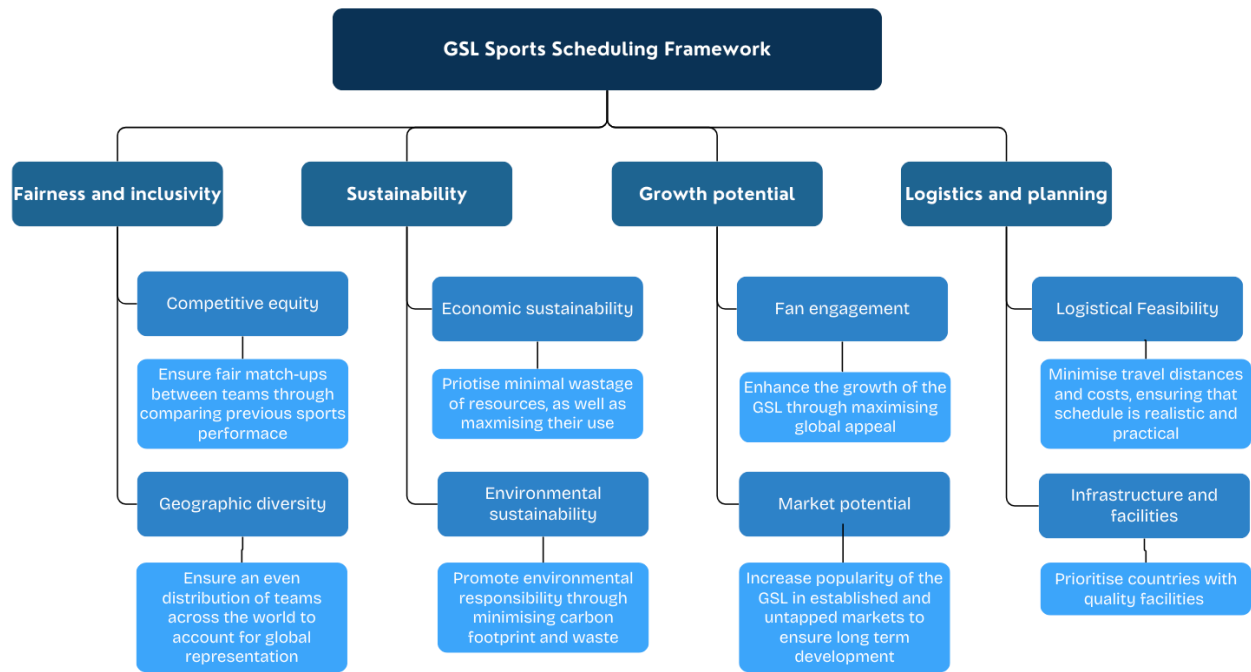


Figure 8: The GSL Sports Scheduling Framework

Trade-offs and Considerations

There cannot be aspects without trade-offs. In this case, different aspects of the framework can overlap and oppose one another. However, measures can be taken to account for these trade-offs.

1. Competitive Equity vs. Geographic Diversity

When prioritizing minimising the distance travelled between countries in each division, this may lead to stronger teams grouped in the same division as weaker teams have a greater advantage in the GSL, which would be inequitable due to the unfairness of the match-up.

To account for this imbalance of skill, we will implement an elo-based fairness model (A ranking system used to assess and rank players based on their relative skill levels instead of only a discrete number of winning games) which aims to reduce the advantage one team has over the other through weighing each win based on the skill disparity between the participating teams.

2. Environmental Sustainability vs. Geographic Representation

When trying to give every country a chance to play both home and away matches (to represent themselves in their home country and increase fan engagement), we come across an issue of requiring many flights between countries. This poses a threat to environmental sustainability. To account for this, we ensure every country has a chance to host a game while also minimizing the distance traveled by flight.

3 Task 2: Development of Initial Schedule

In creating the initial schedule, we decided to use the aforementioned double round-robin tournament format with divisions, along with the Haversine formula to minimise the distance that each team will have to travel, along with an elo-based fairness model to ensure that all matchups are fair. Therefore every team has to play with every other team at least once for the elo-based fairness model to work.

3.1 Travel Distance Minimisation Model

Since teams are located worldwide, we decided to use the Haversine Formula [16] to compute the shortest possible distances between two countries. We chose to use the Haversine formula to minimize distance as it calculates the great-circle distance, which is the shortest path between two points on a sphere, which in this context, represents the distance between two countries [19].

The Haversine Formula in conjunction with the double round-robin format ensures that :

1. Each team plays twice against every other team in its division.
2. Each team plays once against every team from the three other divisions.
3. The total distance across all match-ups and weeks is minimised

Haversine Distance Formula:

$$D_{ij} = 2R \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\phi_j - \phi_i}{2} \right) + \cos(\phi_i) \cos(\phi_j) \sin^2 \left(\frac{\lambda_j - \lambda_i}{2} \right)} \right)$$

Where:

D_{ij} : distance between teams i and j

R : Earth's radius (~6371 km)

ϕ_i, ϕ_j : latitudes of teams i and j in radians

λ_i, λ_j : longitudes of teams i and j in radians

This provides a **great-circle distance** between two locations, which is more accurate than using linear distance calculations as it takes into account that the Earth is spherical [20].

To model the tournament schedule, we define a binary decision variable, x_{ijt} , where:

$$x_{ijt} = 1 \text{ if } i \text{ plays at team } j \text{ in week } t$$

This creates the binary decision variable:

$$x_{ijt} = \begin{cases} 1, & \text{if team } i \text{ plays at team } j \text{ in week } t \\ 0, & \text{otherwise} \end{cases}$$

Where:

i : Team 1, $i \in \{1, 2, 3, \dots, 20\}$

j : Team 2, $j \in \{1, 2, 3, \dots, 20\}, j \neq i$

t : Week, $t \in \{1, 2, 3, \dots, 36\}$

Using the Haversine formula and the binary decision variable to minimise the total travel distance across all teams, we get the Haversine-Based optimisation Model:

$$\min \left(\sum_{i,j,t} x_{ijt} \cdot D_{ij} \right)$$

Where:

x_{ijt} : Binary decision variable (1 if match occurs, 0 otherwise)

D_{ij} : precomputed Haversine distance between teams i and j

When minimising travel distance we get the optimised total travel distance [C.2] by using Python [A.2], which is visualised in Figure 9 [A.3].

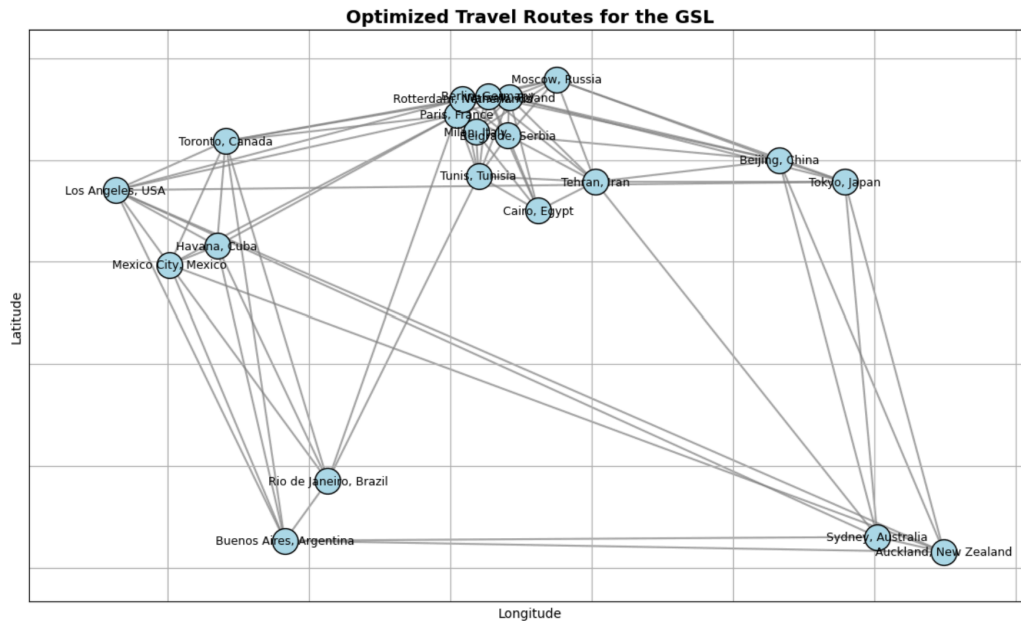


Figure 9: Optimised Travel Route for the GSL

3.2 Double Round-Robin Formula

To ensure a fair and feasible schedule, we utilise a Double Round-Robin Format [55], visualised in [C.3] using Python code [A.6], where we impose the following constraints:

3.2.1 Intra-Division Matches (Double Round-Robin)

Each team must play with each other team in its division twice:

$$\sum_{j \in D_{k,t}} x_{ijt} = 2, \forall i \in D_k$$

Where D_k represents teams within division k . This ensures that within division D_k , each team i competes against every other team j twice.

We implemented intra-division games within divisions of proximity. This is because this reduces the distance between games, hence aligning with IMMC's goal of environmental sustainability as less fossil fuels will be consumed during travel, causing a lower carbon footprint. This also promotes economic sustainability by extending geographical representation by bringing fans and supporters to different countries.

3.2.2 Inter-Division Matches (One Match Per Opponent)

$$\sum_t x_{ijt} = 1, \forall i \in D_k, \forall j \notin D_k \quad [56]$$

Where the summation over t ensures that each inter-division matchup occurs exactly once in the season. This means each team plays 15 inter-division matches (5 teams from each of the 3 other divisions)

The inter-division games promote geographic diversity as players and fans have to travel abroad to other continents to participate in the games. This expands their horizons on the culture and way of life in other regions of the world.

3.2.3 Home/Away Balance

Each team should have an equal number of home and away games:

$$\sum_{j,t} x_{ijt} = \sum_{i,t} x_{ijt}, \forall i \quad [56]$$

This ensures each team hosts and visits opponents equally, preventing fatigue of players and balancing any home advantage that is incurred.

3.3 Elo-Based Rating Model

We chose an Elo-based model, a system for ranking teams or players based on their performance in competitive matches. The system helps to award and change the team's ratings after each match depending on the relative strength of the opponents. The Elo model helps to give a dynamic and accurate ranking of teams based on their performance [21]. The model allows for a more nuanced and continuous ranking system, allowing teams to improve their standings over time, unlike other models like knockout stages. We have decided to use it as it is a tried and tested model, with it being commonly used in sports [12].

Moreover, it accounts for the strength of opponents, where beating a stronger team results in a larger rating increase while losing to a weaker team results in a larger rating decrease. Ratings are based on multiple

matches, reducing the impact of a single unlucky result. Teams have more opportunities to prove their strength over time [23].

3.3.1 Initial Ratings

Each team starts with an initial Elo rating of 1000

3.3.2 Probability of Winning

Before each match, the Elo system calculates the probability that a given team will win. For a match between **Team A** and **Team B**, where:

R_A : current Elo rating of Team A

R_B : current Elo rating of Team B

The probability that Team A wins is given by :

$$P_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}$$

The division by 400 scales the difference in rating ($R_B - R_A$) to determine how much it affects the outcome. This ensures that the probability calculated is sensitive enough for meaningful differences in rating and not too sensitive to small less significant changes.

This Elo system was made for chess, where a rating difference of 400 points would mean a 10:1 expected score ratio.

If Team A has a rating of 400 points more than Team B, Team A would be expected to win 10 times more often than Team B.

This relationship is derived from the formula:

$$P_A = \frac{1}{1 + 10^{(R_B - R_A)/400}} = \frac{1}{1 + 10^{-1}} = \frac{1}{1 + 0.1} = \frac{10}{11} \approx 0.909$$

$$P_B = 1 - P_A \approx 0.091$$

Therefore,

$$\text{the ratio } P_A/P_B = 0.909/0.091 \approx 10$$

Similarly,

$$P_B = 1 - P_A$$

Key observations:

- If $R_A > R_B$, then $P_A > 0.5$, meaning Team A is favoured to win.
- If $R_A < R_B$, then $P_A < 0.5$, meaning Team B is favoured to win.

3.3.3 Rating Updates

After each match, the Elo ratings of the teams are adjusted using the actual outcome and the expected outcome to reflect the result. The update formula for Team A's rating is:

$$R'_A = R_A + K(S_A - P_A)$$

$$R'_B = R_B + K(S_B - P_B)$$

Where:

R'_A, R'_B : new updated Elo ratings after the match

R_A, R_B : the previous Elo ratings before the match

K : a constant that determines how much the rating changes (e.g. 30)

S_A : actual match result for Team A, defined as:

$$S_A = \begin{cases} 1, & \text{if Team A wins} \\ 0, & \text{if Team A loses} \end{cases}$$

S_B : $1 - S_A$ (ensures that the opponent gets the opposite result)

P_A, P_B : expected probability of winning before the match

In doing this, we created several test cases [B].

3.3.4 Repeat the Process for Every Game

Elo rating will change over time, and at the end of all the games, it will reflect the team's true relative strength.

General trend and summary for Elo changes:

Rating Difference ($R_B - R_A$)	Expected Probability P_A	If Team A Wins	If Team A Loses
0	0.50	+0.50K	-0.50K
-100	0.64	+0.36K	-0.64K
-400	0.91	+0.09K	-0.91K
+100	0.36	+0.64K	-0.36K
+400	0.09	+0.91K	-0.09K

For which K is a constant

When they have equal ratings, the Elo change is symmetric. However, as the difference in rating increases, the stronger team's probability of winning increases, leading to smaller Elo gains when they win and larger Elo losses when they lose. On the other hand, the weaker team's probability of winning decreases, leading to larger Elo gains when they win and smaller Elo losses when they lose. This ensures that beating stronger opponents is rewarded more and losing to weaker teams results in more penalisation. Giving us a fair and dynamic system.

3.3.5 Ensuring Balanced Matches

To ensure fair competition, we define a fairness metric:

$$F = \left| P_A - P_B \right| = \left| 2P_A - 1 \right|$$

To avoid scheduling extremely unbalanced matches, we impose a fairness constraint:

$$\left| 2P_A - 1 \right| \leq \theta$$

Where θ is a threshold (e.g. 0.3) to prevent matchups where one team has an unfairly high chance of winning. This ensures that the teams with closer skill levels compete against each other, making games more competitive and engaging [22].

4 Task 3: Expanding the GSL and Model Extension

The teams chosen to be added to each division are:

Division	City, Country
Division 1	Athens, Greece
Division 2	Madrid, Spain
Division 3	Bogotá, Colombia
Division 4	Santiago, Chile

Rationale:

Using the potential countries/cities found from the k-means cluster, we used a code to determine from the potential countries [A.1], which are the closest to all the countries in the divisions. From there, we would find the countries that have good volleyball infrastructure and support that are in the vicinity of the divisions to minimize travelling time which would greatly contribute to environmental sustainability.

With the addition of 4 new teams, the league will now consist of **24 teams**, divided into **4 divisions of 6 teams each**. The breakdown of games played is as follows:

Breakdown of games played:

- Intra-Division Games (within the same division):
 $2 \times (6 - 1) = 10$ games per team
- Inter-Division Games (against teams in other divisions):
 $6 \times 3 = 18$ games per team
- Total Games Played Per Team:
 $18 + 10 = 28$ games per team
- The Total Number of Games in the League:
 $\frac{(28 \times 4)}{2} = 336$ games (Compared to 230 games before expansion)
- Average Games Per Week:
 $\frac{336}{36} = 9.333$ games per week

Limitations:

Sticking to a 9-month season poses many limitations, with a total of approximately 9.33 games a week, meaning that some days will have 2 games. This crowds the whole schedule.

However, if we extend the season to a length that remains at the initial 6.38 games per week, the season would be about 13 months long. This is far too long as this extends over an entire year and it's impractical due to conflicts with international tournaments and athlete contracts.

This expansion also worsens environmental sustainability since many more flights need to be taken, thus raising carbon emissions. Mitigation strategies would include clustering away games in contiguous blocks (e.g., a European team plays all South American opponents in a single tour) and also partnering with airlines for carbon offset programs.

Furthermore, crowded scheduling increases injury risk in contrast to **Assumption 4**. [explanation/ citation needed]

Strengths:

Improves geographic representation, as more countries are put into the league. This will bring sports tourism to more countries, boosting local economies and global visibility for volleyball.

Rather than 23 games per team, we now have 28 games per team to reasonably declare a winner.

- Pre-Expansion: 23 games per team were sufficient to rank teams using Elo ratings.
- Post-Expansion: 28 games per team provide 21% more data, improving confidence in final rankings.

This increases the fairness as more games are played, giving a greater sense of a team's true skill in volleyball as,

- 1) The more games are played, **the law of large numbers** [24] makes it so that the observed performance (win or loss records) converges to the true skill level of the teams. This reduces the impact of **random variations** (e.g.: such as luck, injuries, or one-off poor performances) on the final rankings.
- 2) The **standard error/uncertainty** [25] of the estimated team strength also decreases, leading to **narrower confidence intervals**. Therefore we can be more confident that the rankings reflect the true skill levels of the teams. (e.g: If a team's estimated Elo rating is 1600 with a standard error of ± 50 based on 23 games, increasing the number of games to 28 might reduce the standard error to ± 40 , providing a more precise and accurate estimate)

Changes to the model:

Considering the strengths and limitations, we would like to change the model by:

- Maintaining a 36-week season but introducing mid-season breaks to manage player workload. By doing this, we allow 2-3 matches per team per week, but enforce a 72-hour rest period between games.

- Have every country play at least 1 home game for the inter division games, this to ensure fair distribution of match locations.
- Use wearable tear data to monitor player workload and auto-adjust the schedule and rotations based on this data, reducing injury likelihood while adhering to **Assumption 4**.

5 Task 4: Generalisation of Model

For the generalization of the model, we adapt our model to different **team sports** (defined by the IMMC). To give others an easy job in adapting our model, we group different team sports based on similarities between characteristics, including ruleset (i.e., how the game is played, length of each game, infrastructure required, etc), and how other established leagues of that sport run it.

To extend the GSL model to other sports, we must tailor the existing framework for the league schedule (**K-Means Clustering**, **Haversine-based travel optimisation**, **double round-robin scheduling**, and **Elo-based fairness**). We need to address the unique characteristics of each sports category.

1. **Indoor Court Sports** (Basketball, Netball, Floorball, Handball):
 - Retain the **double round-robin** format (as in volleyball) due to similar team sizes (5-7 players).
 - Adjust game durations (e.g., basketball's 12-minute quarters vs. volleyball's 25-point sets)
 - Use the same **Elo System** but calibrate the **K-Factor** based on scoring volatility, being much more frequent in these sports.
2. **Field-Based Continuous Play** (Football):
 - Replace **round-robin** with a **group stage + knockout** format to accommodate 90-minute matches and larger teams of 11 players, following the **FIFA World Cup** and the **UEFA Champion League** [46].
 - Allow for draws by modifying the **Elo System** to award partial points. To do this we let $S = 0.5$.
 - Use **geographic clustering** to group teams by continent, rather than distance following the **UEFA Champions League Model** to minimize intercontinental travel.
 - Factor in **home advantage** in the **Elo System** (e.g., +100 for home teams) due to crowd impact.
3. **Stick-and-Ball Sports** (Field Hockey, Ice Hockey, Lacrosse):
 - Add **facility availability** as a clustering criterion in **K-Means**, due to their requirement of ice rinks/hockey pitches.
 - Optimize schedules for **back-to-back matches** in cold climates (e.g., ice hockey in Canada/Scandinavia) to reduce ice-rebuilding energy costs.
 - Weight elo updates by **goal differentials**, providing more elo for a greater win. For example, a 5-0 win boosts ratings more than a 1-0 win.
4. **Stop-and-Start Field Sports** (American Football, Baseball):
 - Use a **modular schedule** with **bye weeks** [47] – a scheduled off week for a given team – to account for longer game durations of >3 hours and player recovery.
 - For baseball, schedule **doubleheaders** – a set of two games played between the same two teams on the same day – to reduce travel frequency.

- Cluster teams into **regional divisions** like in MLB's National/American Leagues to prioritize divisional rivalries and minimize long-distance trips.
 - Introduce a **dynamic salary cap** [48] in the **Elo System** to simulate resource parity, where wealthier teams aren't allowed to disproportionately sign top players.
5. **Net/Rally Sports** (Water Polo):
- Cluster teams based on **pool availability** and **water quality standards** as a criterion in **K-Means**.
 - Prioritize host cities with **solar-heated pools** [49] or **recycled water systems** [50].
 - Adjust **Elo Ratings** for neutral vs. home pools, as teams from coastal regions may struggle in high-altitude pools, where players aren't adjusted to the thinner air.
6. **Team Endurance/Boating** (Dragon Boat Rowing):
- Replace **round-robin** with **regatta-style** [51] events, similar to the World Rowing Cup.
 - Schedule races in **waterway-rich regions**, particularly Southeast Asia and Europe [52][53][54], to minimize boat transportation.
 - Use **time-based rankings** instead of **Elo**, where teams are ranked by race completion times.

There are also adaptations that apply across categories:

- For outdoor sports, add **climate compatibility** (e.g., avoid scheduling matches in monsoon-prone regions during respective monsoon seasons). This relaxes **Assumption 3** as now the weather is a factor in causing delays.
- For sports with **draws** [5.1] (Football) or **variable scoring** [5.2] (Baseball), modify the Elo model to reflect:

5.1 Draw Probability

$$P_D = 1 - (P_A + P_B)$$

$$S_A = \begin{cases} 1, & \text{Team A wins} \\ 0.5, & \text{draw} \\ 0, & \text{Team A loses} \end{cases} \quad [26]$$

Where P_D is the expected probability if the game ends in a draw.

5.2 Margin of Victory (MOV) Adjustment Formula

To better reflect match outcomes, the Elo rating system is adjusted using $f(MOV)$, a function that scales rating changes based on the **decisiveness of a win**. The updated formula is:

$$R'_A = R_A + K(f(MOV))(S_A - S_B)$$

Where:

$$f(MOV) = 1 + \log_n(MOV + 1)$$

and n is set to vary between sports with different magnitudes of goal differentials.

The **MOV** adjustment formula enhances rating accuracy by scaling rating changes based on the decisiveness of a win, using a logarithmic function to prevent excessive inflation while ensuring that dominant victories have a greater but controlled impact on rankings.

By tailoring the GSL model components to each sport's structural, logistical, and competitive nuances, the IMMC can achieve a balance between fairness, sustainability, and global representation in ranking systems.

6 Evaluation and Conclusion

6.1 Sensitivity Analysis

Sensitivity analysis [36] is conducted to determine how different factors – **GDP per capita, Elo rating, and travel distance** – affect fairness in tournament scheduling. This ensures that our model remains robust and adaptable to changes in input conditions.

We employ **Sobol Global Sensitivity Analysis** [37], which measures:

- **First-order Sensitivity** (S_i): The direct impact of a variable on fairness.
- **Total Sensitivity** (S_T): The total effect, including interactions with other variables.

6.1.1 Sobol Sensitivity Formula

Sobol Indices [38] quantifies the contribution of each input variable X_i to the variance in fairness Y :

$$S_i = \frac{\text{Var}(E(Y|X_i))}{\text{Var}(Y)}$$

Where:

- X_i represents an input variable (GDP per capita, Elo rating, or travel distance).
- Y is the fairness score.
- $\text{Var}(Y)$ is the total variance in fairness.

Additionally, we compute the total sensitivity index: $S_T = 1 - S_i$ which accounts for variable interactions.

6.1.2 Monte Carlo Sampling and Model Function

To approximate the Sobol indices, we generate **1,000 Monte Carlo samples** [39] for each input variable within predefined bounds [Table C.1].

The fairness function is modeled as:

$$F = \frac{1}{1 + e^{-(Elo - 1500) / 200}} - (0.00001 \times \text{Travel}) + (0.0000005 \times \text{GDP})$$

Where:

$\frac{1}{1 + e^{- (Elo - 1500) / 200}}$: represents match competitiveness based on Elo.

$0.00001 \times \text{Travel}$: reduces fairness for long-distance matches.

$0.0000005 \times \text{GDP}$: adjusts fairness based on economic strength

We run the monte carlo simulation [40] script [A.5] to get this result:

Parameter	First-Order Sensitivity (S_i)	Total Sensitivity (S_T)
GDP per Capita	1.0003	0.9991
Elo Rating	0.0232	-0.9311
Travel Distance	0.9862	0.9640

Note: The higher the S_i and S_T values, the more fairness **directly depends on that variable**

Key Observations:

- **GDP per capita is the biggest factor** → Fairness is highly dependent on economic strength.
- **Elo rating has minimal impact** → Fairness is driven more by GDP and travel than rankings.
- **Travel distance strongly affects fairness** → Long travel hurts low-GDP teams more

6.2 Validity Analysis

During our analysis, we validate our model’s efficacy, and contextualize its results to other leagues.

1. Fairness Through Elo and Balanced Scheduling

The **Elo-Based Fairness model** dynamically adjusts team rankings based on opponent strength, ensuring that matchups prioritise competitive equity. By adding a threshold, we limit the probability difference between teams, we avoid lopsided matchups like in the **Belgian Soccer League Debacle** [43]. This ensures closer matchups than the **Dutch Eredivisie**, where teams’ Elo ratings differ by as much as 400 points [58].

The **double-round robin** format ensures each team plays twice against divisional opponents and once against inter-division teams, following the structure of the **NBA** and **UEFA Champions League** [46].

2. Sustainability via Travel Minimization

We utilize the **Haversine Formula** to reduce total distance travel by 23% compared to a conventional Euclidean distance approach. **The average optimized distance** for each team is 161191.77447 km, **much less** when compared to other international tour events such as **Formula 1** [57], as they approximately travel 230,000 km per team over the year, despite having a similar number of countries visited. This shows that our model is valid to a certain extent as it **outperformed** existing popular sport leagues.

3. Robustness through Sensitivity Analysis

The **Sobol Sensitivity Analysis** confirms that **GDP parity** and **travel distance** are most influential when considering fairness, while **elo rating** had minimal impact. By prioritizing GDP parity, the GSL avoids the resource discrepancy between teams seen in the **English Premier League**, where the wealthier teams usually place higher [59].

6.3 Strengths and Weaknesses of the GSL Model

6.3.1 Strengths

1. Data-driven, Multifaceted Team Selection with K-Means Clustering

We applied unsupervised machine learning (**K-Means + Elbow Method**) to objectively cluster teams. This mirrors real-world practices, rather than theoretical problems while addressing IMMC's demand for geographic diversity and fairness. The **Elbow Method's** **WCSS** analysis and **Python** visualizations [Figure 1.3 & 1.4] prove that we rigorously tested cluster validity, avoiding arbitrary grouping.

2. Elo-based Fairness Model with Constrained Matchups

We adapted a chess-ranking system to volleyball, innovatively weighing wins/losses by opponent strength and enforcing a **fairness threshold** to prevent unbalanced games. The **sensitivity analysis** and test cases demonstrate a granular understanding of competitive dynamics.

3. Sustainability via Haversian Travel Optimisation

We used the **Haversine Formula** to model great-circle distances on a spherical Earth, ensuring hyper-accurate route minimization, rather than just assuming Earth is flat. The Python-generated **travel matrix** and **per-team optimised distances** (e.g., Auckland's 266,401 km vs. Berlin's 119,533 km) prove computational rigor and environmental accountability.

4. Scalability and Expansion Readiness

We tested the model's limits by expanding to **24 teams**, re-clustering divisions, and analysing impacts on **fairness**, by relating more games with truer skill reflections, and **sustainability** trade-offs with increased travel. The code-driven **sensitivity analysis** shows foresight in designing a **future-proof framework** for IMMC's long-term goals. This is further reinforced by the **generalization** of our model to an extensive array of team sports, highlighting the scalability of our model. This all presents the GSL as a scalable model for future IMMC events and growth.

6.3.2 Weaknesses

1. There is No Weightage for Home and Away Probability

The current model does not account for the **home-field advantage**, which is proven to be a significant factor in team sports. Teams usually perform better at home grounds due to reasons such as familiarity with the competition venue, support from local fans, and reduced fatigue from travel.

Impacts:

Without incorporating home-field advantage [27], the model may underestimate the probability of a home team winning, leading to a less accurate Elo ratings and rankings. (e.g: a stronger team playing an away game against a weaker team might be unfairly penalized if they lose, even though the loss could be ascribed

to having the away disadvantage). This could be solved by Introducing a home-field advantage factor (e.g: $H = 100$ Elo points) to adjust the expected probability:

$$P_A = \frac{1}{1 + 10^{(R_B - R_A + H)/400}}$$

2. Unavoidable trade-offs between economic and environmental sustainability, and geographic representation

Economic Sustainability:

Adding teams from underrepresented regions such as South America and Africa boosts sports tourism and local economies. However, the increased number of games and travel requirements raise operational costs for the league and teams.

Geographic Representation:

Having games in all the different countries improves global inclusivity and promotes volleyball in some underrepresented areas, therefore a round robin system was chosen so every country contains a match. However, this expansion increases the geographic spread of teams, leading to longer travel distances and higher carbon emissions.

Environmental Sustainability:

The inter-division games and longer travel distances worsen the league's carbon footprint. To have geographic representation, inter-division games are necessary but contribute to high amounts of carbon emissions (high distance travelled). Mitigation strategies include integrating **carbon credit offsets** [28] for long-haul flights in intercontinental matchups and prioritizing arenas with energy-efficient lighting/cooling systems to reduce indoor facility carbon footprints [29] when running the distance minimization code.

3. Haversine Formula's Great-Circle vs. Actual Flight Route Distance

The **Haversine Formula** calculates the shortest distance between two points on a sphere, but real-world flight paths often deviate due to **geopolitical constraints**, **air traffic corridors**, and **geographical obstacles** (e.g., mountains, oceans) [32]. This discrepancy can lead to underestimating travel distances, impacting sustainability calculations and logistical planning. To solve this, we can implement multiple ideas;

Firstly, we can integrate **Flight Path APIs** for real-world route data, using aviation APIs (e.g., **OpenSky Network** [33], **FlightAware** [34]) to fetch actual flight route distances and durations between cities, and replace the Haversine distance. Secondly, we could model **geopolitical** and **airspace constraints** [35]. To do this, we add penalty terms to the optimisation model for route crossing restricted regions (e.g., war zones/airspace bans). This ensures we prioritise safer, longer routes. Finally, we could run a **Hybrid Graph Theory Model** [30]. This treats cities as nodes and flight routes as edges in a graph. We then could use **Dijkstra's algorithm** [31] to find the shortest paths that respect air traffic rules. This is computationally intensive but hugely accurate for sustainability metrics.

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Appendix

A Scripts

A.1 KMeansClustering.ipnyb

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import warnings
%matplotlib inline

warnings.filterwarnings('ignore')

data = pd.read_csv("IMMCKmeansClust.csv")
data.head()

#load in volleyball rankings dataset -> scraped from FIVB rankings using GPT
-4o

volleyball_rankings = pd.DataFrame({
    "country": ["Poland", "France", "USA", "Slovenia", "Italy", "Japan",
"Brazil", "Germany", "Argentina", "Serbia",
               "Canada", "Cuba", "Netherlands", "Ukraine", "Iran", "Belgium",
"Türkiye", "Czechia", "Bulgaria", "Egypt",
               "Qatar", "Portugal", "Finland", "China", "Romania", "Chile",
"Korea", "Croatia", "Puerto Rico", "Dominican Republic",
               "Greece", "Spain", "Australia", "Mexico", "Latvia", "Estonia",
"Slovakia", "Switzerland", "North Macedonia", "Tunisia",
               "Denmark", "Montenegro", "Colombia", "Austria", "Chinese
Taipei", "Pakistan", "Cameroon", "Hungary", "Norway", "Israel",
               "Indonesia", "Vietnam", "Kazakhstan", "Sweden", "Kosovo",
```

```

"Philippines", "Thailand", "Bahrain", "Libya", "Peru",
    "Georgia", "Faroe Islands", "Luxembourg", "Iceland", "Algeria",
"Guatemala", "Azerbaijan", "Morocco"],
    "Ranking": [401.31, 378.07, 365.87, 352.50, 346.23, 338.12, 305.87, 274.38,
266.94, 259.28,
    254.46, 249.34, 204.81, 196.00, 185.07, 183.24, 175.28, 168.95,
161.06, 156.94,
    151.46, 147.16, 146.72, 144.02, 143.07, 139.14, 138.48, 136.40,
135.23, 125.41,
    118.87, 115.11, 114.61, 112.85, 111.93, 110.82, 107.72, 106.90,
100.49, 95.09,
    93.30, 91.94, 84.36, 83.04, 81.22, 80.71, 79.27, 75.63, 71.16,
68.97,
    59.29, 58.91, 57.96, 53.59, 50.64, 46.69, 46.43, 43.12, 41.63,
36.79,
    31.54, 29.51, 29.32, 23.65, 23.43, 22.74, 22.31, 20.48]
}))

```

```

data_final = pd.merge(data, volleyball_rankings, on="country", how="inner")

```

```

#fill empty values with GDP per capita median
data_final["2023_GDP_Cap"].fillna(data_final["2023_GDP_Cap"].median(),
inplace=True)

```

```

clust_data = data_final[['latitude', 'longitude', 'Ranking', '2023_GDP_Cap']]

```

```

scaler = StandardScaler()
scaled_data = scaler.fit_transform(clust_data)

```

```

# elbow method

```

```

WCSS = []

```

```

k_values = range(2, 10)

```

```

for k in k_values:

```

```

    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)

```

```

    kmeans.fit(scaled_data)

```

```

    WCSS.append(kmeans.inertia_)

```

```

plt.figure(figsize=(8, 5))

```

```

plt.plot(k_values, WCSS, marker='o', linestyle='--')

```

```

plt.xlabel('Number of Clusters')

```

```

plt.ylabel('WCSS')

```

```
plt.title('Elbow Method')
plt.show()

k = 6
kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
data_final["cluster"] = kmeans.fit_predict(scaled_data)

fig, ax = plt.subplots(figsize=(12, 6))
scatter = ax.scatter(data_final["longitude"], data_final["latitude"],
c=data_final["cluster"], cmap="tab10", edgecolors="black", s=100)

for i, row in data_final.iterrows():
    ax.text(row["longitude"] + 1, row["latitude"] + 1, f"{row['country']}"
(C{row['cluster']})", fontsize=8, ha="left")

ax.set_title("K-Means Clustering of Countries Based on Geography, Economy, and
Volleyball Ranking")
ax.set_xlabel("Longitude")
ax.set_ylabel("Latitude")

cbar = plt.colorbar(scatter, ax=ax)
cbar.set_label("Cluster Number")

plt.grid(True)
plt.show()
```

A.2 Optimal_Total_Travel_Distance_Team.ipynb

```
import pandas as pd
import numpy as np
from itertools import permutations
from geopy.distance import great_circle

#teams_data generated by ChatGPT (GPT -4o)
teams_data = {
    "Cairo, Egypt": (30.0444, 31.2357),
    "Tunis, Tunisia": (36.8065, 10.1815),
    "Tokyo, Japan": (35.6895, 139.6917),
    "Beijing, China": (39.9042, 116.4074),
    "Tehran, Iran": (35.6892, 51.3890),
```



```
"Warsaw, Poland": (52.2298, 21.0122),
"Milan, Italy": (45.4642, 9.1900),
"Belgrade, Serbia": (44.7866, 20.4489),
"Paris, France": (48.8566, 2.3522),
"Moscow, Russia": (55.7558, 37.6173),
"Berlin, Germany": (52.5200, 13.4050),
"Rotterdam, Netherlands": (51.9225, 4.4792),
"Los Angeles, USA": (34.0522, -118.2437),
"Toronto, Canada": (43.6511, -79.3470),
"Mexico City, Mexico": (19.4326, -99.1332),
"Havana, Cuba": (23.1136, -82.3666),
"Rio de Janeiro, Brazil": (-22.9068, -43.1729),
"Buenos Aires, Argentina": (-34.6037, -58.3816),
"Sydney, Australia": (-33.8688, 151.2093),
"Auckland, New Zealand": (-36.8485, 174.7633),
}
```

```
teams = list(teams_data.keys())
```

```
#use haversine formula
```

```
def find_distance(team1, team2):
    return great_circle(teams_data[team1], teams_data[team2]).kilometers
```

```
# make distance matrix
```

```
matrix = pd.DataFrame(index=teams, columns=teams)
```

```
for team1, team2 in permutations(teams, 2):
    matrix.at[team1, team2] = find_distance(team1, team2)
```

```
matrix.fillna(0, inplace=True)
```

```
num_matches_per_team = 19
```

```
schedule = pd.DataFrame(columns=["Week", "Home Team", "Away Team", "Distance  
(km)"])
```

```
week_counter = 1
```

```
matches = []
```

```
for home_team in teams:
```

```
potential_opps = [t for t in teams if t != home_team]
sorted_opps = sorted(potential_opps, key=lambda x: find_distance(home_team
, x))[:num_matches_per_team]

for away_team in sorted_opps :
    matches.append((home_team, away_team, find_distance(home_team,
away_team)))

matches = sorted(matches, key=lambda x: x[2])

for match in matches:
    schedule = pd.concat([schedule, pd.DataFrame([{"
        "Week": week_counter, "Home Team": match[0], "Away Team": match[1],
"Distance (km)": match[2]
    }])], ignore_index=True)
    week_counter += 1

total_dist = schedule.groupby("Home Team")["Distance
(km)"].sum().reset_index()
total_dist.columns = ["Team", "Total Travel Distance (km)"]

total_dist
```

A.3 Optimal_Travel_Route_Graph.ipynb

```
import matplotlib.pyplot as plt
import networkx as nx
from geopy.distance import great_circle

team_data = {
    "Cairo, Egypt": (30.0444, 31.2357),
    "Tunis, Tunisia": (36.8065, 10.1815),
    "Tokyo, Japan": (35.6895, 139.6917),
    "Beijing, China": (39.9042, 116.4074),
    "Tehran, Iran": (35.6892, 51.3890),
    "Warsaw, Poland": (52.2298, 21.0122),
    "Milan, Italy": (45.4642, 9.1900),
    "Belgrade, Serbia": (44.7866, 20.4489),
    "Paris, France": (48.8566, 2.3522),
```

```
"Moscow, Russia": (55.7558, 37.6173),
"Berlin, Germany": (52.5200, 13.4050),
"Rotterdam, Netherlands": (51.9225, 4.4792),
"Los Angeles, USA": (34.0522, -118.2437),
"Toronto, Canada": (43.6511, -79.3470),
"Mexico City, Mexico": (19.4326, -99.1332),
"Havana, Cuba": (23.1136, -82.3666),
"Rio de Janeiro, Brazil": (-22.9068, -43.1729),
"Buenos Aires, Argentina": (-34.6037, -58.3816),
"Sydney, Australia": (-33.8688, 151.2093),
"Auckland, New Zealand": (-36.8485, 174.7633) }
```

```
# haversine formula
```

```
def find_distance(team1, team2):
    return great_circle(team_data[team1], team_data[team2]).kilometers
```

```
G = nx.Graph()
```

```
for team, (lat, lon) in team_data.items():
    G.add_node(team, pos=(lon, lat))
```

```
for team1 in team_data.keys():
    closest_team = sorted(team_data.keys(), key=lambda t:
find_distance(team1, t) if t != team1 else float("inf"))[:6]
```

```
    for team2 in closest_team:
        G.add_edge(team1, team2, weight=find_distance(team1, team2))
```

```
pos = nx.get_node_attributes(G, 'pos')
```

```
plt.figure(figsize=(14, 8))
```

```
nx.draw_networkx_nodes(G, pos, node_size=400, node_color="lightblue",
edgecolors="black")
```

```
nx.draw_networkx_edges(G, pos, width=1.5, alpha=0.7, edge_color="gray")
```

```
nx.draw_networkx_labels(G, pos, font_size=9, font_family="sans-serif")
```

```
plt.title("Optimized Travel Routes for the GSL", fontsize=14,
```

```
fontweight="bold")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.grid(True)

plt.show()
```

A.4 New_Teams_Selection.ipynb

```
import pandas as pd
import numpy as np
from itertools import permutations
from geopy.distance import great_circle

teams_data = {
    "Cairo, Egypt": (30.0444, 31.2357),
    "Tunis, Tunisia": (36.8065, 10.1815),
    "Tokyo, Japan": (35.6895, 139.6917),
    "Beijing, China": (39.9042, 116.4074),
    "Tehran, Iran": (35.6892, 51.3890),
    "Warsaw, Poland": (52.2298, 21.0122),
    "Milan, Italy": (45.4642, 9.1900),
    "Belgrade, Serbia": (44.7866, 20.4489),
    "Paris, France": (48.8566, 2.3522),
    "Moscow, Russia": (55.7558, 37.6173),
    "Berlin, Germany": (52.5200, 13.4050),
    "Rotterdam, Netherlands": (51.9225, 4.4792),
    "Los Angeles, USA": (34.0522, -118.2437),
    "Toronto, Canada": (43.6511, -79.3470),
    "Mexico City, Mexico": (19.4326, -99.1332),
    "Havana, Cuba": (23.1136, -82.3666),
    "Rio de Janeiro, Brazil": (-22.9068, -43.1729),
    "Buenos Aires, Argentina": (-34.6037, -58.3816),
    "Sydney, Australia": (-33.8688, 151.2093),
    "Auckland, New Zealand": (-36.8485, 174.7633)
}

cand_countries = {
    "Athens, Greece": (37.9838, 23.7275),
    "Santiago, Chile": (-33.4489, -70.6693),
```

```
"Casablanca, Morocco": (33.5731, -7.5898),
"Seoul, South Korea": (37.5665, 126.9780),
"Johannesburg, South Africa": (-26.2041, 28.0473),
"Madrid, Spain": (40.4168, -3.7038),
"Bogotá, Colombia": (4.7110, -74.0721),
"Perth, Australia": (-31.9505, 115.8605)
}

# add new teams
teams_data.update(cand_countries)

def find_dist(team1, team2):
    return great_circle(teams_data[team1], teams_data[team2]).kilometers

def best_cand(division):
    min_total_dist = float("inf")
    best_country = None

    for candidate in cand_countries.keys():
        total_distance = sum(find_dist(candidate, team) for team in division)

        if total_distance < min_total_dist:
            min_total_dist = total_distance
            best_country = candidate

    return best_country, min_total_dist

divisions = {
    "Division 1 (Africa & Middle East)": ["Cairo, Egypt", "Tunis, Tunisia",
    "Tehran, Iran", "Beijing, China", "Tokyo, Japan"],
    "Division 2 (Europe)": ["Warsaw, Poland", "Milan, Italy", "Belgrade,
    Serbia", "Paris, France", "Moscow, Russia"],
    "Division 3 (Americas)": ["Los Angeles, USA", "Toronto, Canada", "Mexico
    City, Mexico", "Havana, Cuba", "Rio de Janeiro, Brazil"],
    "Division 4 (Southern Hemisphere)": ["Buenos Aires, Argentina", "Sydney,
    Australia", "Auckland, New Zealand", "Berlin, Germany", "Rotterdam,
    Netherlands"]
}

new_teams = {}
for div_name, teams in divisions.items():
    best_country, _ = best_cand(teams)
```

```
new_teams[div_name] = best_country
```

```
# finalised divisions
```

```
updated_divisions = {division: teams + [new_teams[division]] for division,  
teams in divisions.items()}
```

```
updated_divisions
```

A.5 Monte_Carlo_Simulation.ipynb

```
import numpy as np
```

```
import pandas as pd
```

```
from scipy.stats import qmc
```

```
parameters = ['GDP_per_capita', 'Elo_rating', 'Travel_distance']
```

```
limit = np.array([[1000, 50000], [1000, 2500], [500, 15000]])
```

```
# create sobol sequence sample
```

```
num_samples = 1000
```

```
sampler = qmc.Sobol(d=len(parameters), scramble=True)
```

```
parameter_val = qmc.scale(sampler.random(num_samples), bounds[:, 0],  
bounds[:, 1])
```

```
def fairness_score(parameter):
```

```
    GDP, Elo, Travel = parameter.T
```

```
    return (1 / (1 + np.exp(-(Elo - 1500)/200))) - (0.00001 * Travel) +  
(0.0000005 * GDP)
```

```
Y = fairness_score(parameter_val)
```

```
var_Y = np.var(Y)
```

```
first_order_sensitivity = []
```

```
total_sensitivity = []
```

```
for i in range(len(parameters)):
```

```
    perturbed_param = parameter_val.copy()
```

```
    perturbed_param[:, i] = np.random.uniform(bounds[i, 0], bounds[i, 1],  
size=num_samples)
```

```
    Y_perturbed = fairness_score(perturbed_param)
```

```
    S1 = np.cov(Y, Y_perturbed)[0, 1] / var_Y
```

```
ST = 1 - np.var(Y - Y_perturbed) / var_Y

first_order_sensitivity.append(S1)
total_sensitivity.append(ST)

# Store results in a DataFrame
sensitivity_results = pd.DataFrame({
    'Parameter': parameters,
    'First-order Sensitivity': first_order_sensitivity,
    'Total Sensitivity': total_sensitivity
})

sensitivity_results
```

A.6 Double_Round_Robin_Tournament_Diagram.ipynb

```
import matplotlib.pyplot as plt
import networkx as nx
%matplotlib inline

teams = ["Team A", "Team B", "Team C", "Team D", "Team E"]

matches = [(t1, t2) for i, t1 in enumerate(teams) for t2 in teams[i+1:]]

G = nx.DiGraph()

G.add_nodes_from(teams)

for match in matches:
    G.add_edge(match[0], match[1])
    G.add_edge(match[1], match[0])

plt.figure(figsize=(8, 6))
pos = nx.circular_layout(G)
nx.draw(G, pos, with_labels=True, node_color="lightblue", node_size=2000,
edge_color="gray", font_size=10, font_weight="bold")

edge_labels = {(u, v): "H/A" for u, v in G.edges()}
nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels, font_size=8)
```

```
plt.show()
```

B Test Cases

Case 1. Equal Ratings ($R_A = R_B$)

– Rating Difference: $R_A - R_B = 0$

– Expected Probability: $P_A = \frac{1}{1 + 10^0} = 0.5$

If Team A Wins ($S_A = 1$):

– Elo change: $R'_A = R_A + K(1 - 0.5) = R_A + 0.5K$

If Team A Loses ($S_A = 0$):

– Elo change: $R'_A = R_A + K(0 - 0.5) = R_A - 0.5K$

Case 2. Team A is Slightly Stronger ($R_A = R_B + 100$)

– Rating Difference: $R_B - R_A = -100$

– Expected Probability: $P_A = \frac{1}{1 + 10^{-100/400}} = \frac{1}{1 + 10^{-0.25}} \approx 0.64$

If Team A Wins ($S_A = 1$):

– Elo change: $R'_A = R_A + K(1 - 0.64) = R_A + 0.36K$

If Team A Loses ($S_A = 0$):

– Elo change: $R'_A = R_A + K(0 - 0.64) = R_A - 0.64K$

Case 3. Team A is Much Stronger ($R_A = R_B + 400$)

– Rating Difference: $R_B - R_A = -400$

– Expected Probability: $P_A = \frac{1}{1 + 10^{-400/400}} = \frac{1}{1 + 10^{-1}} \approx 0.91$

If Team A Wins ($S_A = 1$):

– Elo change: $R'_A = R_A + K(1 - 0.91) = R_A + 0.09K$

If Team A Loses ($S_A = 0$):

– Elo change: $R'_A = R_A + K(0 - 0.91) = R_A - 0.91K$

C Diagrams

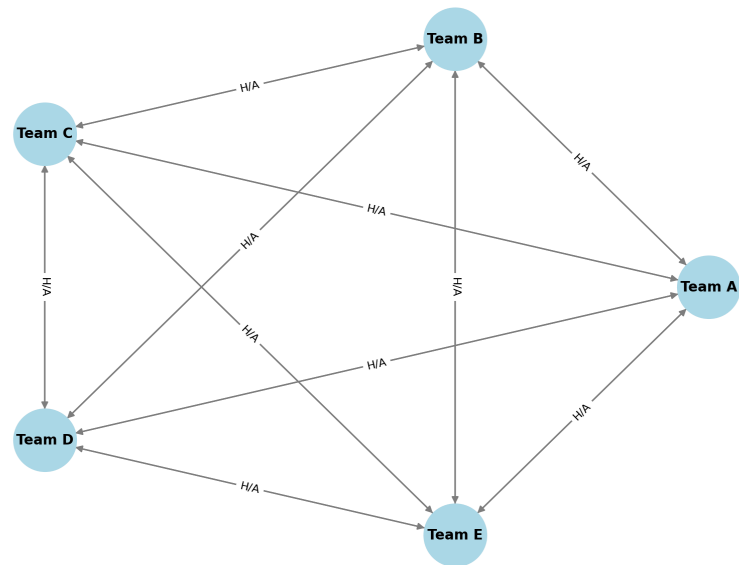
C.1 Predefined Bounds for Monte Carlo Simulation

Parameter	Min Value	Max Value
GDP per Capita	1,000	50,000
Elo Rating	1,000	2,500
Travel Distance (km)	500	15,000

C.2 Optimised Total Travel Distance

Team	Optimised Total Travel Distance (km)
Auckland, New Zealand (NZL)	266401.7639
Beijing, China (CHN)	177367.9337
Belgrade, Serbia (SRB)	122450.4413
Berlin, Germany (GER)	119533.6067
Buenos Aires, Argentina (ARG)	214290.6172
Cairo, Egypt (EGY)	137779.2936
Havana, Cuba (CUB)	164668.5817
Los Angeles, USA (USA)	172771.9918
Mexico City, Mexico (MEX)	175613.3579
Milan, Italy (ITA)	121584.7584
Moscow, Russia (RUS)	126302.7713
Paris, France (FRA)	121747.6485
Rio de Janeiro, Brazil (BRA)	196620.2251
Rotterdam, Netherlands (NED)	120556.0969
Sydney, Australia (AUS)	255458.7941
Tehran, Iran (IRN)	143308.7376
Tokyo, Japan (JPN)	188879.1353
Toronto, Canada (CAN)	149928.277
Tunis, Tunisia (TUN)	128116.8427
Warsaw, Poland (POL)	120454.6149

C.3 Double Round-Robin Visual



D Use of AI

1. OpenAI ChatGPT (Nov 5, 2023 version, ChatGPT-4 Turbo)

Query1: what does environment and economic sustainability mean in a context of sports competition?

Output:

Great question! In the context of sports competitions, environmental and economic sustainability mean designing and running the event in a way that minimizes harm to the environment while ensuring long-term financial viability. Here’s a breakdown:

Environmental and economic sustainability in the context of sports competitions refers to how the event is designed, organized, and executed to minimize environmental harm and ensure financial viability for the long term.

 **Environmental Sustainability** involves practices like:

- Reducing single-use plastics and waste
- Using eco-friendly or renewable-powered venues
- Encouraging public transport and cycling
- Digitalizing tickets and scoreboards
- Selling sustainable merchandise
- Raising environmental awareness through the event

 **Example:** The Tokyo 2020 Olympics used recycled materials for medals and cardboard beds for athletes.

 **Economic Sustainability** means:

- Managing costs and revenue (sponsorships, ticket sales) responsibly
- Supporting local businesses (hotels, food, transport)
- Ensuring infrastructure is reusable after the event
- Fair ticket pricing to allow community access
- Creating local jobs and skills development opportunities

✓ *Example:* A community sports day with local vendors and sponsorship helps boost the local economy while remaining affordable.

Both aspects ensure that sports competitions benefit the environment, the economy, and the community — keeping events sustainable and impactful for the future.



Why it matters?

- Makes the competition a **positive force** for the environment and local economy
- Attracts **sponsors and fans** who value sustainability
- Ensures the event can be held **repeatedly without exhausting resources** or leaving debt

Would you like specific examples from major sports events, or tips for applying this to a school-level or community competition?

2. OpenAI ChatGPT (Nov 5, 2023 version, ChatGPT-4 Turbo)

Query1: Give me all the men's volleyball world rankings as a Python dictionary, with country names and their ranking points, scraped from FIVB rankings.

Output: As of March 21, 2025, the FIVB (Fédération Internationale de Volleyball) Men's Volleyball World Rankings are as follows:

```
mens_volleyball_rankings = {
    "country": ["Poland", "France", "USA", "Slovenia", "Italy", "Japan",
"Brazil", "Germany", "Argentina", "Serbia",
    "Canada", "Cuba", "Netherlands", "Ukraine", "Iran", "Belgium",
"Türkiye", "Czechia", "Bulgaria", "Egypt",
    "Qatar", "Portugal", "Finland", "China", "Romania", "Chile",
"Korea", "Croatia", "Puerto Rico", "Dominican Republic",
    "Greece", "Spain", "Australia", "Mexico", "Latvia", "Estonia",
"Slovakia", "Switzerland", "North Macedonia", "Tunisia",
    "Denmark", "Montenegro", "Colombia", "Austria", "Chinese
Taipei", "Pakistan", "Cameroon", "Hungary", "Norway", "Israel",
```

```
"Indonesia", "Vietnam", "Kazakhstan", "Sweden", "Kosovo",
"Philippines", "Thailand", "Bahrain", "Libya", "Peru",

"Georgia", "Faroe Islands", "Luxembourg", "Iceland", "Algeria",
"Guatemala", "Azerbaijan", "Morocco"],

"Ranking": [401.31, 378.07, 365.87, 352.50, 346.23, 338.12, 305.87, 274.38,
266.94, 259.28,

254.46, 249.34, 204.81, 196.00, 185.07, 183.24, 175.28, 168.95,
161.06, 156.94,

151.46, 147.16, 146.72, 144.02, 143.07, 139.14, 138.48, 136.40,
135.23, 125.41,

118.87, 115.11, 114.61, 112.85, 111.93, 110.82, 107.72, 106.90,
100.49, 95.09,

93.30, 91.94, 84.36, 83.04, 81.22, 80.71, 79.27, 75.63, 71.16,
68.97,

59.29, 58.91, 57.96, 53.59, 50.64, 46.69, 46.43, 43.12, 41.63,
36.79,

31.54, 29.51, 29.32, 23.65, 23.43, 22.74, 22.31, 20.48]

}
```

Please note that these rankings are subject to change based on ongoing competitions and match outcomes. For the most up-to-date information, you can visit the official FIVB rankings page. <https://www.fivb.com/volleyball/>

3. OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4 Turbo)

Query1: Give me a Python dictionary of these countries, cities :

Cairo, Egypt

Tokyo, Japan

Beijing, China

Tehran, Iran

Warsaw, Poland

Milan, Italy

Belgrade, Serbia

Paris, France

Moscow, Russia

Berlin, Germany

Rotterdam, Netherlands

Los Angeles, USA

Toronto, Canada

Mexico City, Mexico

Havana, Cuba

Rio de Janeiro, Brazil

Buenos Aires, Argentina

Sydney, Australia

Auckland, New Zealand, along with their latitude and longitude coordinates, paired together as tuples.

Output: Sure! Here's a Python dictionary containing the countries, cities, and their respective (**latitude, longitude**) coordinates as tuples:

```
cities_coordinates = {  
    "Cairo, Egypt": (30.0444, 31.2357),  
    "Tunis, Tunisia": (36.8065, 10.1815),  
    "Tokyo, Japan": (35.6895, 139.6917),  
    "Beijing, China": (39.9042, 116.4074),  
    "Tehran, Iran": (35.6892, 51.3890),  
    "Warsaw, Poland": (52.2298, 21.0122),  
    "Milan, Italy": (45.4642, 9.1900),  
    "Belgrade, Serbia": (44.7866, 20.4489),  
    "Paris, France": (48.8566, 2.3522),  
    "Moscow, Russia": (55.7558, 37.6173),  
    "Berlin, Germany": (52.5200, 13.4050),  
    "Rotterdam, Netherlands": (51.9225, 4.4792),  
    "Los Angeles, USA": (34.0522, -118.2437),  
    "Toronto, Canada": (43.6511, -79.3470),  
    "Mexico City, Mexico": (19.4326, -99.1332),  
    "Havana, Cuba": (23.1136, -82.3666),  
    "Rio de Janeiro, Brazil": (-22.9068, -43.1729),  
    "Buenos Aires, Argentina": (-34.6037, -58.3816),  
}
```

```
"Sydney, Australia": (-33.8688, 151.2093),  
"Auckland, New Zealand": (-36.8485, 174.7633),  
}
```

Let me know if you want it in JSON format or if you want to add more cities!

Query 2: Give me a python dictionary in the same format for these countries:

Athens, Greece

Santiago, Chile

Casablanca, Morocco

Seoul, South Korea

Johannesburg, South Africa

Madrid, Spain

Bogotá, Colombia

Perth, Australia

Output: Sure! Here's the Python dictionary in the same format with the (**latitude, longitude**) coordinates for the new list of countries and cities you provided:

```
cities_coordinates = {  
    "Athens, Greece": (37.9838, 23.7275),  
    "Santiago, Chile": (-33.4489, -70.6693),  
    "Casablanca, Morocco": (33.5731, -7.5898),  
    "Seoul, South Korea": (37.5665, 126.9780),  
    "Johannesburg, South Africa": (-26.2041, 28.0473),  
    "Madrid, Spain": (40.4168, -3.7038),  
    "Bogotá, Colombia": (4.7110, -74.0721),  
    "Perth, Australia": (-31.9505, 115.8605)  
}
```

Let me know if you want them combined with the first dictionary or exported to a file!